Logistic Regression:

linear regression

When we have normally distributed data we use linear regression with

$$\left[\begin{array}{l} \text{mean} = 0 \\ \sigma^2 = \text{variance} \\ y = \beta_0 + \beta_1 x + \varepsilon \\ E[Y/x] = \beta_0 + \beta_1 x \end{array}\right]$$

- σ^2 is not a function of x
  if Y is not Normal

=> If the data has discrete distribution then we cannot use linear regression

if Y is Bernoulli

$$E[Y|x] = P$$

$$= \frac{1}{1+\exp[-(\beta_0+\beta_1 x)]}$$

if Y is Binomial

$$E[Y/x] = np$$

$$= \frac{n}{1 + \exp[-(\beta_0 + \beta_1 x)]}$$

Binary Response (Logistic Regression)

$$E[Y/x] = P = \frac{1}{1 + \exp[-\beta_0 + \beta_1 x]}$$

$$1 + \exp[-(\beta_0 + \beta_1 x)] = \frac{1}{P}$$

$$\exp[-(\beta_0 + \beta_1 x)] = \frac{1-P}{P}$$

$$\exp(\beta_0 + \beta_1 x) = \frac{P}{1-P} \Rightarrow \text{Odds of success}$$

$$\boxed{\beta_0 + \beta_1 x = \ln\left(\frac{P}{1-P}\right)}$$

$$\frac{\exp(\beta_0 + \beta_1 x_{12})}{\exp(\beta_0 + \beta_1 x_{11})} = \frac{[P/(1-P)]_2}{[P/(1-P)]_1} \rightarrow \text{Odds Ratio}$$

Example 12.13: The data set in Table 12.16 will be used to illustrate the use of logistic regression to analyze a single-agent quantal bioassay of a toxicity experiment. The results show the effect of different doses of nicotine on the common fruit fly.

Table 12.16: Data Set for Example 12.13

x Concentration (grams/100 cc)	n_i Number of Insects	y Number Killed	Percent Killed
0.10	47	8	17.0
0.15	53	14	26.4
0.20	55	24	43.6
0.30	52	32	61.5
0.50	46	38	82.6
0.70	54	50	92.6
0.95	52	50	96.2

The purpose of the experiment was to arrive at an appropriate model relating probability of "kill" to concentration. In addition, the analyst sought the so-called **effective dose** (ED), that is, the concentration of nicotine that results in a certain probability. Of particular interest was the ED_{50}, the concentration that produces a 0.5 probability of "insect kill."

B0 and b1 will be given

$$E[Y] = n_i p_i = \frac{n_i}{1+\exp[-(\beta_0+\beta_1 x_i)]}$$

$$\hat{p} = \frac{1}{1+\exp[-1.7361+6.2954x]}$$

ED_{50} → Effective Dosage produces $P = 0.5$

when $P = 0.5$

$$\ln\left(\frac{P}{1-P}\right) = 0$$

$$\beta_0 + \beta_1 x = 0$$

$$x = \frac{-\beta_0}{\beta_1} = 0.276.$$